

Requirement Analysis

Data Flow Diagram & User Stories

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether it's through targeted promotional campaigns, adjustments in product formulations, or personalized

recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Data Flow Diagram

Step 1 — Raw Source: cosmetics.csv, the raw cosmetics product extract (1,472 rows, 11 columns)

Step 2 — Data Cleaning & Preprocessing: Python (Pandas) used to standardize brand-name casing, replace scrape-artifact ingredient placeholders (e.g. “No Info”, “Visit the [Brand] boutique”) with missing values, convert a Rank of 0 to missing (unrated), and rename the skin-type columns for clarity

Step 3 — Cleaned Dataset: cosmetics_cleaned.xlsx (1,472 rows, 11 columns) loaded into Tableau

Step 4 — Tableau Data Source: Excel workbook connected to Tableau; Price and Rank fields verified as numbers, Label/Brand/Name/Ingredients verified as text

Step 5 — Data Reshaping: the five skin-type indicator columns (Combination_Skin, Dry_Skin, Normal_Skin, Oily_Skin, Sensitive_Skin) pivoted into two long-format fields, Skin_Type and Suitable, expanding the working table from 1,472 to 7,360 rows

Step 6 — Visualization Layer: four chart sheets built — Avg Price by Category (bar), Price vs Rating (scatter), Top Brands by Category (bar, top 10), and Skin Type Coverage (heatmap)

Step 7 — Dashboard Layer: a single interactive dashboard (1366×768) assembled from the four chart sheets, with the Avg Price by Category chart set as a cross-chart filter action and a searchable Brand filter control added

Step 8 — Publishing: workbook titled “Cosmetic Insights: Trends & Consumer Behavior” and saved to Tableau Public; shareable URL generated for team and stakeholder access

User Stories

User Type	Epic	User Story	Priority	Sprint
Consumer (Priya)	Data Visualization	As a skincare consumer, I want to see how price compares to rating so I can judge value before buying	High	Sprint 2
Consumer (Priya)	Dashboard	As a skincare consumer, I want a single dashboard showing price, rating, and skin-type suitability together	High	Sprint 2
Retail Manager (Meera)	Data Visualization	As a retail category manager, I want to see average price by category so I can plan pricing and stock	High	Sprint 2
Retail Manager (Meera)	Dashboard	As a retail category manager, I want top brands and category pricing visible together to plan assortment	High	Sprint 2
Brand Analyst (Arjun)	Data Visualization	As a brand analyst, I want to see which brands rank highest by rating so I can benchmark my brand	Medium	Sprint 2
Brand Analyst (Arjun)	Dashboard	As a brand analyst, I want a filterable brand view to compare my brand against competitors	Medium	Sprint 2
All Users	Data Preparation	As a stakeholder, I want the underlying data cleaned of placeholder text and invalid ratings so insights are trustworthy	High	Sprint 1
All Users	Web Integration	As a stakeholder, I want to access the dashboard from any browser without installing Tableau or logging in	High	Sprint 3
All Users	Documentation	As a stakeholder, I want documentation summarizing the dashboard's purpose and key insights	Medium	Sprint 3